

ANAVI TPM 2.0

DATASHEET

Overview

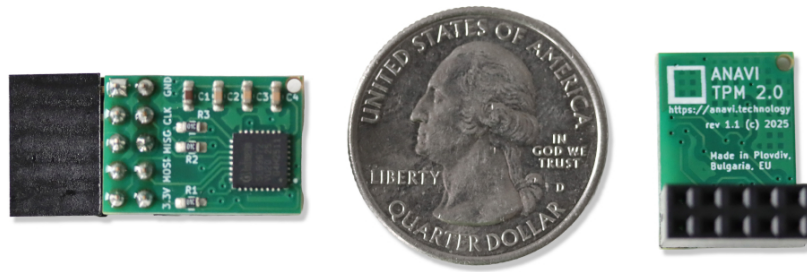


Figure 1: ANAVI TPM 2.0

ANAVI TPM 2.0 is an open-source hardware add-on that provides a **Trusted Platform Module (TPM) 2.0** for Raspberry Pi single board computers. It is designed to deliver hardware-based security, including secure key storage, cryptographic operations, and platform integrity.

The board is built around the **Infineon OPTIGA™ TPM SLB 9672**, a certified TPM 2.0 security controller compliant with modern standards and cryptographic requirements.

The **OPTIGA™ TPM SLB 9672** is a high-end, standardized TPM 2.0 security controller designed for PCs, embedded systems, and IoT devices. It provides strong cryptographic capabilities and future-proof security features including post-quantum protection.

Key Features

ANAVI TPM 2.0

- Open-source hardware design

- Certified by the Open Source Hardware Association (OSHW) under UID BG000152
- Compatible with Raspberry Pi single board computers via GPIO (2x5 header)
- Available in:
 - Horizontal version ANAVI-SPI-TPM2-H
 - Vertical (90°) version ANAVI-SPI-TPM2-V
- Supports SPI communication interface
- Functions as:
 - Secure key storage
 - Cryptographic processor
 - True Hardware Random Number Generator (TRNG)
- Made in Plovdiv, Bulgaria, EU

ANAVI TPM 2.0 Variants

Variant	Mounting	Part Number
ANAVI TPM 2.0 Horizontal	Horizontal	ANAVI-SPI-TPM2-H
ANAVI TPM 2.0 Vertical	Vertical	ANAVI-SPI-TPM2-V



Figure 2: ANAVI TPM 2.0

Security Features & Compliance

Infineon OPTIGA TPM SLB 9672 used in ANAVI TPM 2.0 is:

- TCG TPM 2.0 Library Specification Rev. 1.59
- High confidence level based on Common Criteria and FIPS certification
- High-performance cryptographic engine
- RSA, ECC, AES, SHA, NIST RNG SP800-90A/B, 3GPIO
- Firmware update capability with PQC protection
- Full personalization with 3 endorsement keys (EK) and 3 EK certificates (RSA-2048, ECC NIST P256, ECC NIST P384)

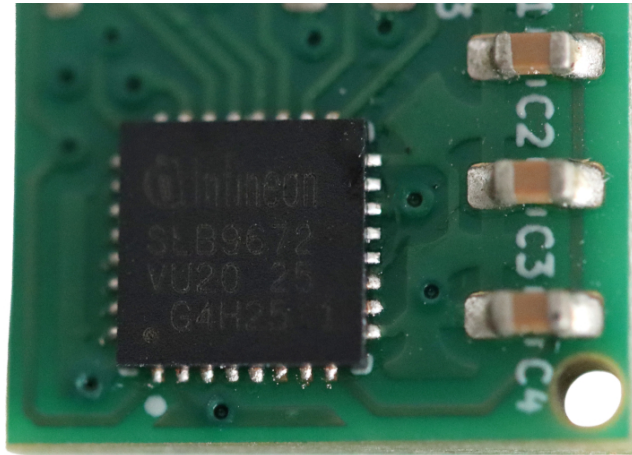


Figure 3: ANAVI TPM 2.0

Electrical & Interface Characteristics

Parameter	Value
Interface	SPI
Supply Voltage	3.3 V
Raspberry Pi Pins	2x5 pins: 17 to 26
PCB Dimensions	18.4 x 12.9 mm (0.72 x 0.51 inches)
Temperature Range	-20°C to +85°C

Software & Compatibility

- Native support in Linux distributions, including Raspberry Pi OS
 - Works with standard TPM drivers and tools
 - Compatible with existing TPM device tree overlays in Raspberry Pi OS
 - Supported by development tools and example code
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Applications

- IoT and edge devices
 - Secure Raspberry Pi projects
 - Industrial automation
 - Networking equipment (routers, gateways)
 - Servers and embedded systems
 - Smart buildings and home automation
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Mechanical & Integration

- Form factor: compact add-on board
- Connector: 2x5 pin female header (2.54 mm)
- Mounting options: horizontal or vertical
- Compatible with all Raspberry Pi single board computers

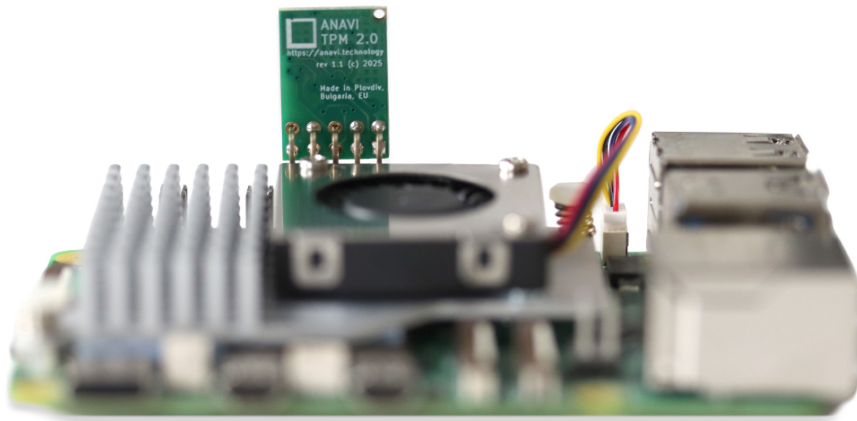


Figure 4: ANAVI TPM 2.0